University Of Colombo, Sri Lanka

UCSC

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination – Semester I – UCSC AY20 [held in October 2023]

SCS1205 – Computer Systems

(Two (2) Hours)

Answer ALL questions

233

Number of Pages = 15 Number of Questions = 4

To be completed by the candidate

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Important Instructions to candidates:

1. The medium of instructions and questions is **English**.
2. Note that questions appear on both sides of the paper.
3. If a page or part of this paper is not printed, please inform the supervisor immediately.
4. Write your index number on every page of the Question paper.
5. Answer **ALL** questions.
6. There are **04** questions in **15** pages.
7. All **04** questions carry **equal marks**.
8. Write your answers and index number clearly and **readable** manner.
9. In questions where justifications, reasons, or steps of calculations are specifically asked, the student must provide the same in the answer. **Failure to provide** the above as required in the question will result in that question or part being awarded **zero marks**.
10. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
11. Calculators are **NOT allowed**.

For Examiner's use only	
Question No	**Marks**
1	
2	
3	
4	
Total	

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Question 1

A. Transistors are of paramount importance to *Central Processing Units* (CPUs) for many fundamental reasons. State whether the following statements are **True** or **False**.

[05 Marks]

#	Statement	True/ False
i.	Transistors serve as the basic building blocks of digital logic circuits within CPUs. They can be used to represent binary values (0 and 1) and perform logical operations (AND, OR, NOT).	
ii.	Transistors are used in the creation of digital memory elements, such as flip-flops and registers, which are essential for storing and manipulating data temporarily during computation.	
iii.	Transistors have become increasingly miniaturized over the years, due to advancements in semiconductor manufacturing processes. This miniaturization and integration enable CPUs to perform complex calculations and execute complex instructions within a small, power-efficient form factor.	
iv.	As CPU manufacturers increase transistor counts, CPUs become more powerful and capable of handling demanding computational tasks. More transistors allow for greater parallelism and improved processing capabilities.	
v.	Smaller transistors are inherently more energy efficient. However, they require more power to switch states and open or close pathways to allow data to move or stop as needed.	

B. What is the **two's complement** 8-bit binary integer representation of the decimal value `-3` ?

[02 Marks]

C. Convert the decimal number `11.6875` to its binary form and write the answer in the cage below. Show your calculations and steps clearly.

[03 Marks]

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D. What is the **decimal** representation of the **hexadecimal** number **3A4C**? Show your intermediate steps clearly.

[03 Marks]

E. Assume the binary number **10.0101** is represented in a **normalized** floating-point representation with a *sign bit*, a 3-bits excess-3 represented *exponent*, and a 4-bits *significand*. What is the error that would occur due to this representation? Show your calculations and steps clearly.

[05 Marks]

F. In *IEEE 754 single precision* floating point number representation, 1 bit is allocated for the sign bit, 8 bits are allocated for the exponent in *excess-127*, and 23 bits are allocated for the significand (mantissa). What is the corresponding IEEE 754 single precision floating point representation for the decimal number -17.375? Write your answer in the relevant cages of the following layout diagram. Show your calculations and steps clearly.

[07 Marks]

Sign 1 bit	Exponent 8 bits	Significand (Mantissa) 23 bits

Question 2

A. A group of students are designing a digital circuit system where it receives an instruction encoded in a 16-bit word as the input.

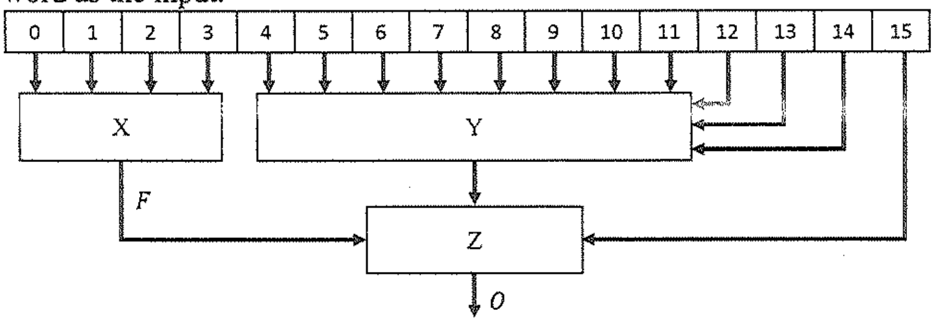


Figure 2.1 – A block diagram of the proposed digital system.

As it is shown in Figure 2.1 there is a component X in this digital system which extracts the first **four (04)** bits of the instruction as the input for the component X. The students have identified that, in any given instance, the first four bits of the instruction cannot hold **all HIGH** (all 1s) values or **all LOW** (all 0s) values. The output *F* of the component X should produce HIGH value, only when the ***Binary Coded Decimal*** value of the input is a prime number.

Assuming the four inputs for the component X as a, b, c, and d, complete the output columns of the following truth table by following the standard notation. Indicate state **`HIGH`** by `1`, Boolean logic state **`LOW`** by `0`, and ***Don't care conditions*** by ×.

[04 Marks]

a	b	c	d	F
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

B. Transfer the values in the truth table above in the question 2.A into the *Karnaugh Map* given below and derive the most simplified Boolean logic expression in the form of **sum of products**. Show your groups clearly.

[04 Marks]

	$\bar{a}.\bar{b}$	$\bar{a}.b$	$a.b$	$a.\bar{b}$
$\bar{c}.\bar{d}$				
$\bar{c}.d$				
$c.d$				
$c.\bar{d}$				

C. The component Y in the Figure 2.1 selects a one input among bits 4 to 11 and forwards the selected input to the output line Q, based on the select values set by the bits 12, 13, and 14. State the **precise name** of the combinational circuit unit that can be used for the component Y? Justify your answer with a brief explanation.

[03 Marks]

D. The component Z in the Figure 2.1 is supposed to produce HIGH value in its output, if and only if its inputs are having **odd** number of HIGH values. What is the logic gate (*i.e.* – a three input single logic gate) that can be used as the component Z? Justify your answer with a brief explanation.

[03 Marks]

E. Complete the following truth table for ***Full Subtractor***. m and s represent *minuend* and the *subtrahend* respectively. B_{in} is the *Borrow-In* bit of a typical full subtractor. $Diff$ and B_{out} represent the *Difference* and the *Borrow-Out* of the subtractor respectively.

[05 Marks]

Inputs			***Outputs***	
m	s	B_{in}	$Diff$	B_{out}
0	*0*	*0*		
0	*0*	*1*		
0	*1*	*0*		
0	*1*	*1*		
1	*0*	*0*		
1	*0*	*1*		
1	*1*	*0*		
1	*1*	*1*		

F. Consider the sequential logic circuit given in Figure 2.2, which consist of a ***negative edge triggered*** J-K Flip Flop, a **positive edge triggered** J-K Flip Flop, and an AND gate. The clock input is given as '**Clk**' in the diagram.

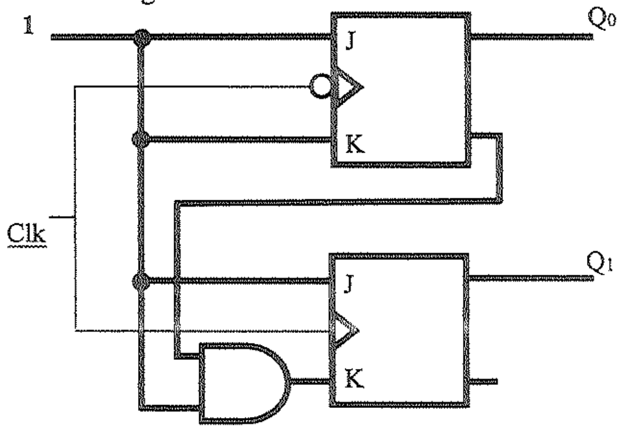
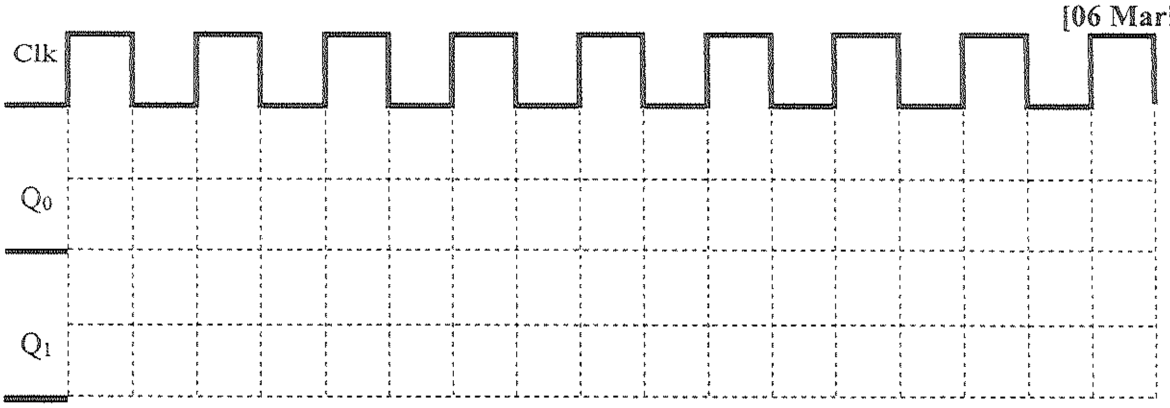


Figure 2.2 – The circuit diagram of the sequential logic circuit.

Complete the timing diagram given below for the above sequential logic circuit. Assume both Q_0 and Q_1 are logic value zero (0) at the beginning.

[06 Marks]



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Question 3

A. Identify the most appropriate CPU component that is responsible for the following tasks listed below. State the CPU component in the corresponding space given in the following table.

[05 Marks]

Task	CPU Component
i. Coordinates and manages the execution of instructions, control data flow between different CPU components, and ensure the proper sequencing of operations in accordance with the program's instructions.	
ii. Keeps track of the memory address of the next instruction to be fetched and executed, ensuring the sequential execution of program instructions.	
iii. Provides high-speed access to frequently used data and instructions, reducing the time it takes to fetch data from main memory.	
iv. Performs arithmetic operations and logical operations on data, enabling the CPU to carry out mathematical calculations and make logical decisions.	
v. Holds the current instruction being executed, allowing the CPU to decode and execute it.	

B. Identify the specific phase (Fetch, Decode, or Execute) of the *Fetch-Decode-Execute cycle* that corresponds to each of the following tasks within a program's execution. State the phase in the corresponding space given in the following table.

[03 Marks]

Task	Phase
i. Determines the operands involved in the instruction, including their locations and addressing modes.	
ii. Go to the main memory and bring the instruction found at the address in the *Memory Address Register*, placing this instruction in the *Instruction Register*.	
iii. Performs the actual operation specified by the OPCODE of the instruction.	
iv. Control signals are generated based on the opcode and operands, preparing the CPU for the execution phase.	
v. Copy the contents of the *Program Counter* to the *Memory Address Register*.	
vi. Interpret the group of bits that has been allocated for the Opcode to determine the operation.	

C. Consider a CPU with the following instruction set architecture:

- **ADD**: Takes two operands from registers and adds them, storing the result in a destination register.
- **SUB**: Subtracts the second operand from the first operand, storing the result in a destination register. *Cycles Per Instruction* (CPI) for SUB is 3.
- **MUL**: Multiplies the two operands, storing the result in a destination register. CPI for MUL is 4.

Assume that the CPU has a ***clock cycle time*** of ***10ns*** and all instructions access registers directly and do not involve memory operations. Given a program that performs the following arithmetic computation. The purpose of each instruction is given as a comment next to the instruction.

```
ADD R1, R2, R3     // R1 = R2 + R3
SUB R4, R1, R5     // R4 = R1 - R5
MUL R6, R4, R7     // R6 = R4 * R7
ADD R4, R1, R6     // R4 = R1 + R6
```

It has been identified that the total execution time for the above program is ***110ns***. Based on the provided information, what is the CPI value of the **ADD** instruction? Show your calculations and steps clearly.

[04 Marks]

D. Consider a machine architecture with an instruction set defined as follows.

LD R A	*Load the register* R *with the content of memory cell with the address* A
LI R I	*Load the register* R *with the value* I *in hexadecimal*
LR R1 R2	*Load the register* R1 *with the content of the register* R2
ST R A	*Store the content of register* R *to the memory cell whose address is* A
ADD R0 R1 R2	*Add the two integers in registers* R1 *and* R2 *and place the result in* R0.
SUB R0 R1 R2	*Subtract the integer in register* R2 *from the integer in register* R1 *and place the result in* R0.
JMP R A	*Jump to the instruction located in the memory cell with the address* A *if the bit pattern in* R *is equal to* R0
HLT	*Halt the execution*

An instruction layout in this machine architecture is **16 bits** long and **4 bits** of an instruction are allocated for **OPCODE**. Memory addresses are **8 bits** long and a register identifier is also **4 bits** long. The following program is executed in this machine. Leftmost column in this set of instructions is the memory addresses that they are stored.

0x40.	LI R1 0x0A
0x42.	LI R2 0xF6
0x44.	LI R3 0x00
0x46.	LI R4 0xFF
0x48.	LI R5 0x01
0x4A.	LI R6 0xFF
0x4C.	ADD R0 R1 R2
0x4E.	JMP R6 0x5C
0x50.	ADD R7 R1 R4
0x52.	LR R1 R7
0x54.	ADD R8 R3 R5
0x56.	LR R3 R8
0x58.	SUB R6 R1 R3
0x5A.	JMP R0 0x4E
0x5C.	ST R1 0x90
0x5E.	ST R3 0x92
0x60.	HLT

i. Assuming negative values are in ***two's complement*** representation, what is the equivalent **Decimal value** that has been loaded to the registers **R1, R2, R3, R4, R5,** and **R6** respectively by the following instruction in the program?

[03 Mark]

Instruction	The Loaded Decimal Value
• LI R1 0x0A	
• LI R2 0xF6	
• LI R3 0x00	
• LI R4 0xFF	
• LI R5 0x01	
• LI R6 0xFF	

ii. If the memory system in this machine is **byte addressable**, what is the numerical value by which the *Program Counter* (PC) is increased in this CPU? **Justify** your answer.

[03 Marks]

iii. List all the values assigned to register **R6 in the order they are assigned** during the program's execution. State the values in **decimal** and briefly state your reasoning.

[03 Marks]

iv. What are the values stored in memory addresses `0x90` and `0x92` at the time of CPU executing the final instruction `HLT` of this program? **Justify** your answer with a brief explanation.

[04 Marks]

- `0x90 =`
- `0x92 =`

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Question 4

A. Explain briefly how *spatial* and *temporal* localities are utilized in the following code segment.

[04 Marks]

```
int sum = 0;
for (int i = 0; i < array.length; i++) {
    sum += array[i];
}
```

B. In a hypothetical computer system with a 16 Bytes of *set associative cache* and a 1024 Bytes of main memory, the CPU refers sixteen (16) addresses in a sequence as given in the following table. The cache line size is two (02) Bytes, and the memory is byte addressable. All the addresses are in binary representation and data values are just symbols.

Sequence No.	Address	Associated Data
1	0000 1010	A
2	0000 1011	B
3	0010 1100	C
4	0001 0010	D
5	0000 1000	E
6	0010 1111	F
7	0000 1001	G
8	0001 0001	H
9	0010 1101	I
10	0001 0000	J
11	0001 0011	K
12	0010 1110	L
13	0011 1010	M
14	0011 1100	N
15	0011 1000	O
16	0011 0011	P

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i. Assuming the cache system employs **First-In-First-Out** (FIFO) cache replacement policy, fill in the cache layout diagram given below by showing the *Tag*, *Data*, and *Valid Bit* values immediately **after referring** the 16^{th} address in the above address sequence. Keep a cell empty if it is unaffected by these address references.

[08 Marks]

Tag	Index	Offset	Data	Valid Bit
	00	0		
		1		
	00	0		
		1		
	01	0		
		1		
	01	0		
		1		
	10	0		
		1		
	10	0		
		1		
	11	0		
		1		
	11	0		
		1		

ii. If the cache memory given in the above diagram in Question **4.B.i** is assumed as a *K-Way Set Associative* cache, identify the most appropriate number for *K* that corresponds to the cache layout in the diagram. Justify your answers with reasoning.

Note that a Direct Mapped cache can be seen as a *K-Way Set Associative* cache when $K = 1$.

[02 Marks]

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iii. Immediately after referring the above 16 addresses, the CPU refers the following **four** (04) addresses in a sequence. State whether each address reference would result a *cache miss* or *cache hit*. **Justify** your answer with reasons for each case.

- 0011 0010
- 1111 0010
- 0011 0011
- 0011 1010

[08 Marks]

0011 0010 -

1111 0010 -

0011 0011 -

0011 1010 -

C. Consider a computer system with a cache memory that has a cache hit time of 2 *nanoseconds* and a cache miss penalty of 20 *nanoseconds*. If the cache hit rate is 95%, calculate the *Effective Access Time* (EAT) for this cache memory system by showing your steps clearly.

[03 Marks]

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